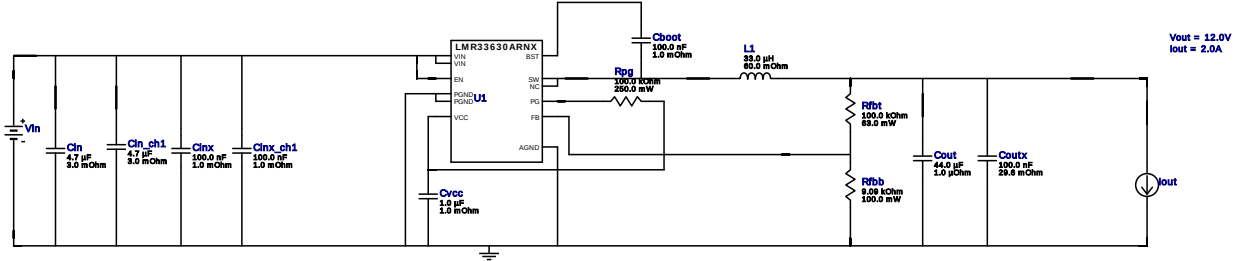


WEBENCH® Design Report

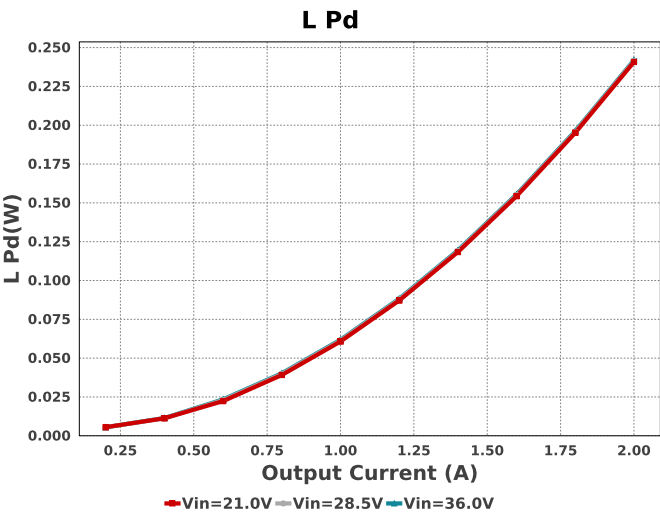
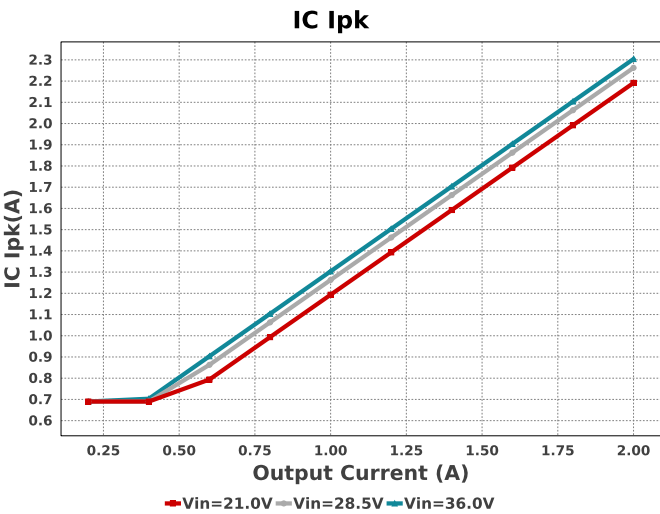
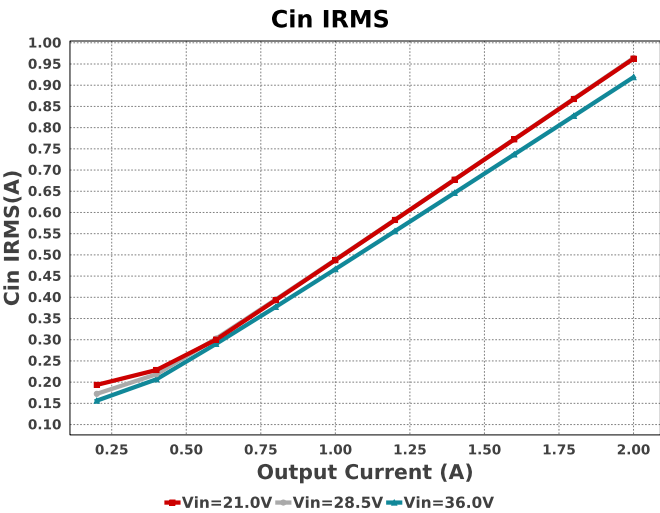
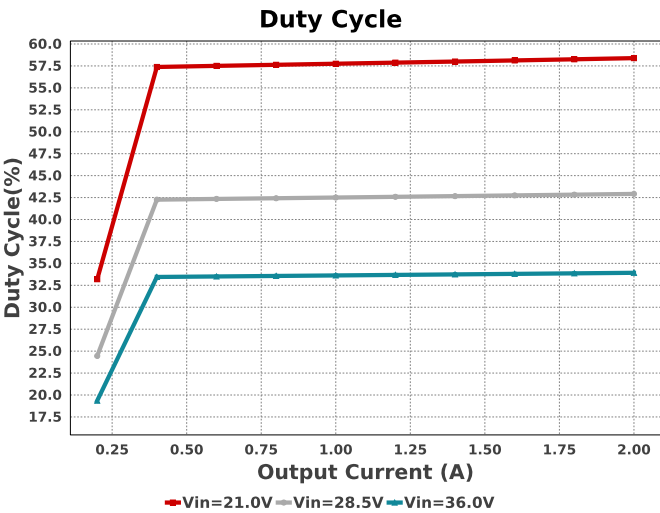
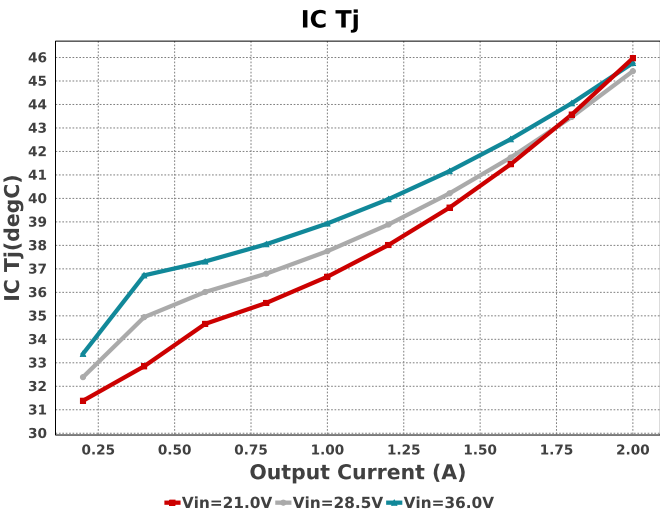
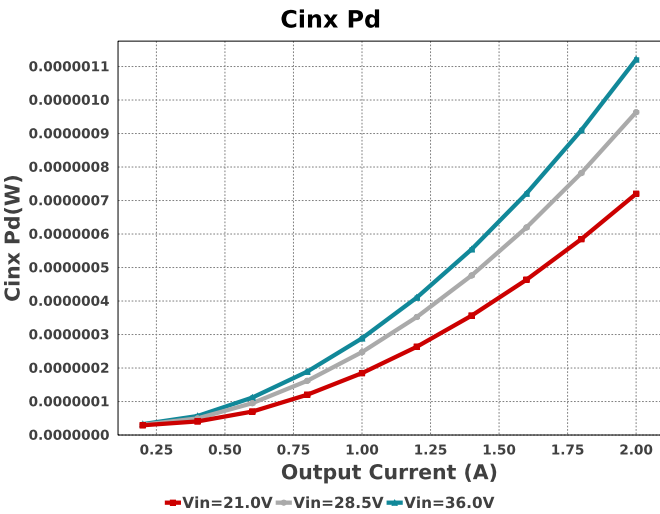
Design : 36 LMR33630ARNXR
LMR33630ARNXR 21V-36V to 12.00V @ 2A

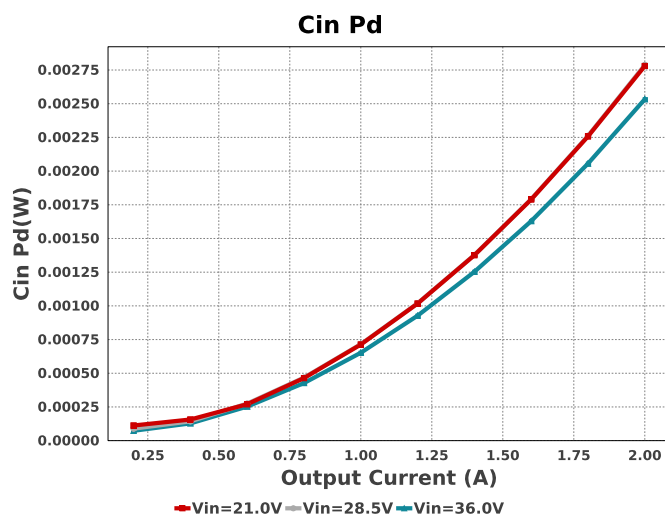
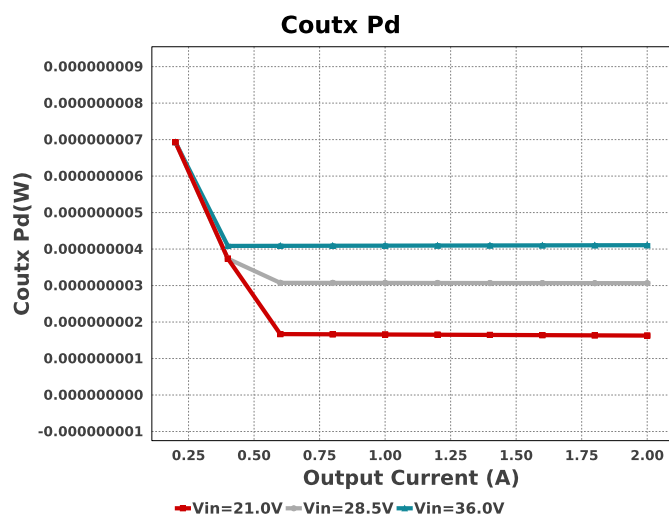
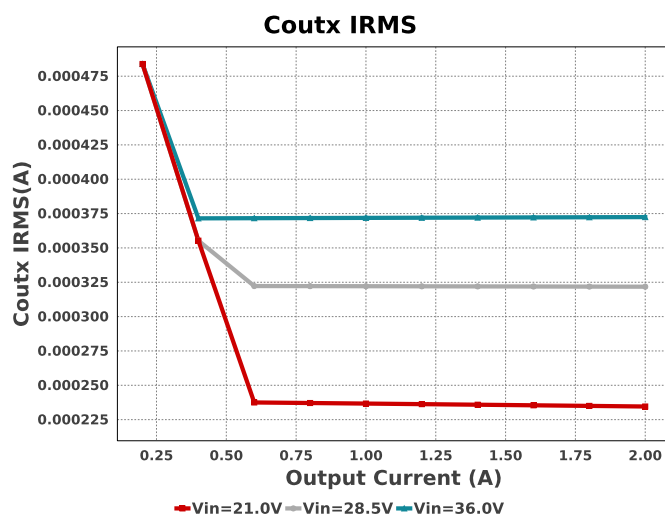
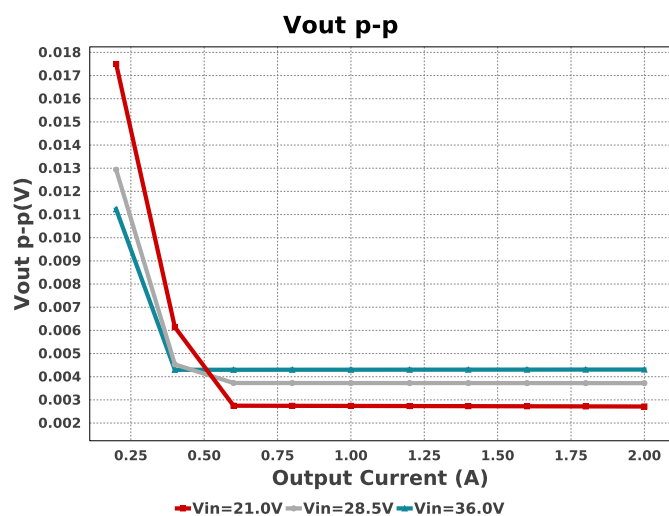
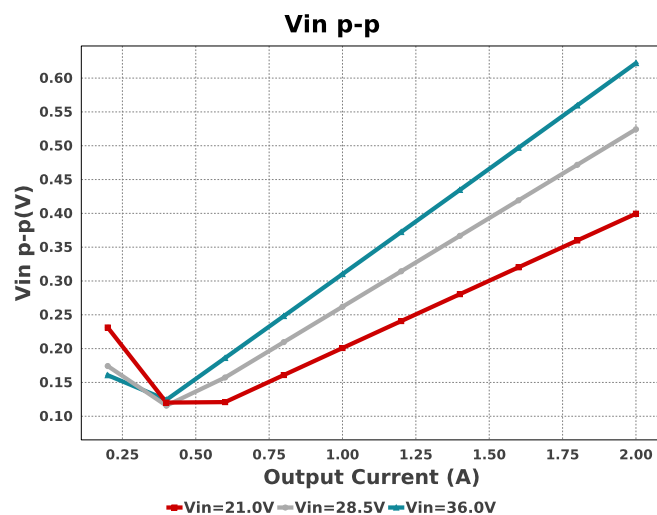
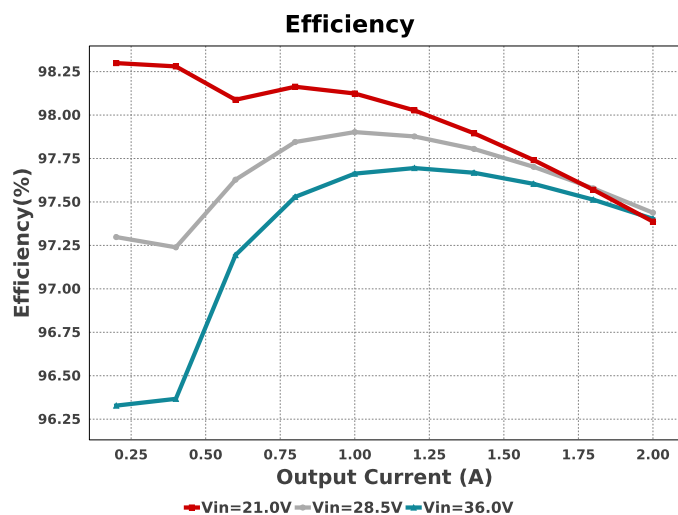


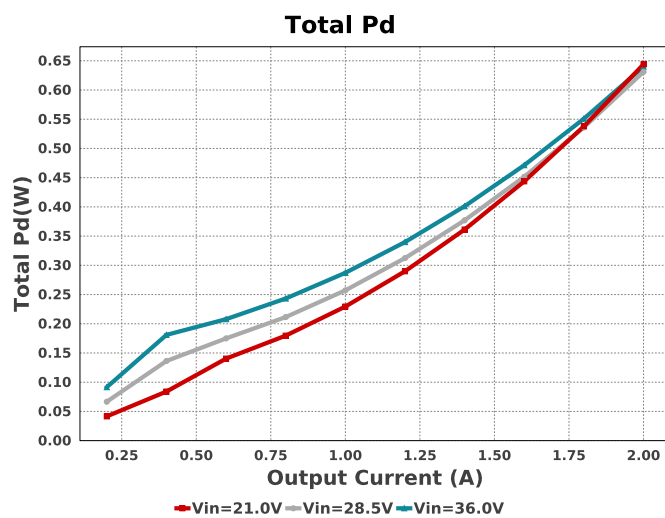
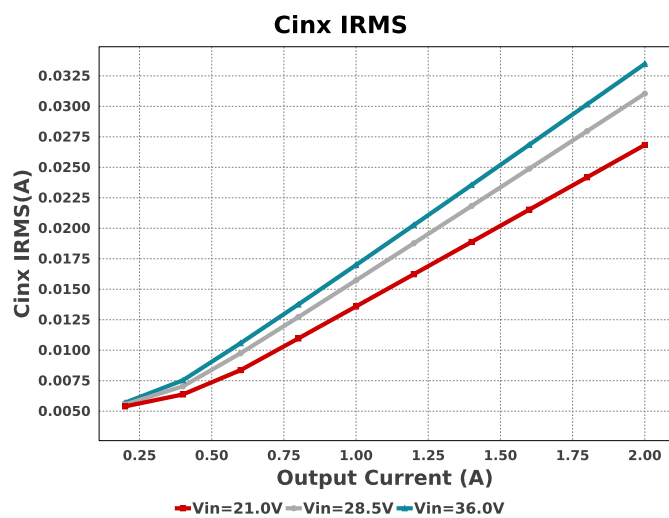
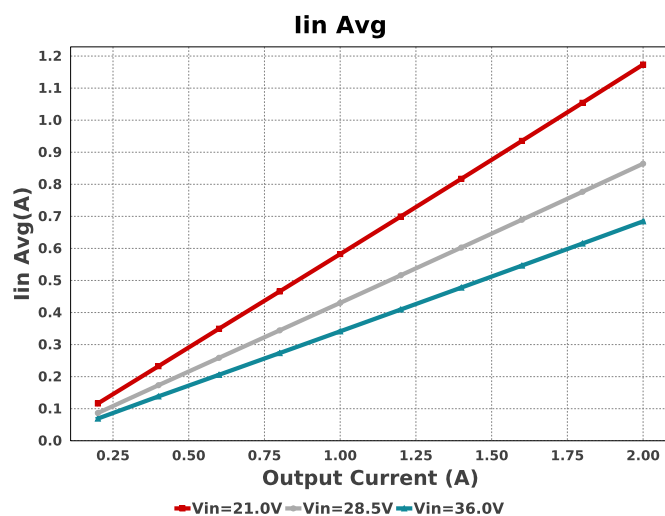
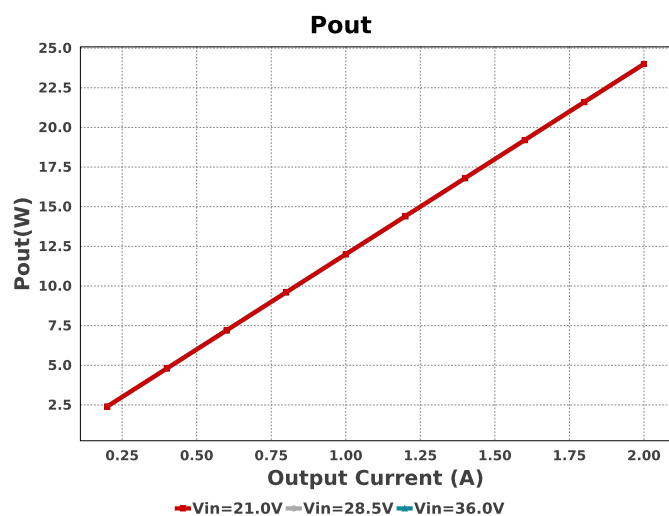
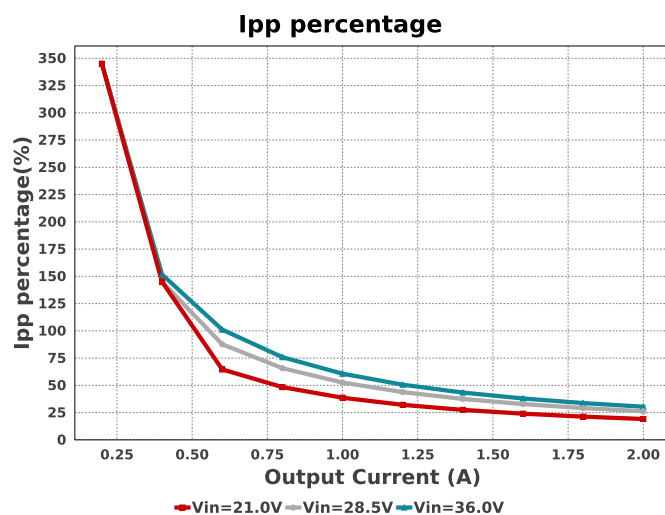
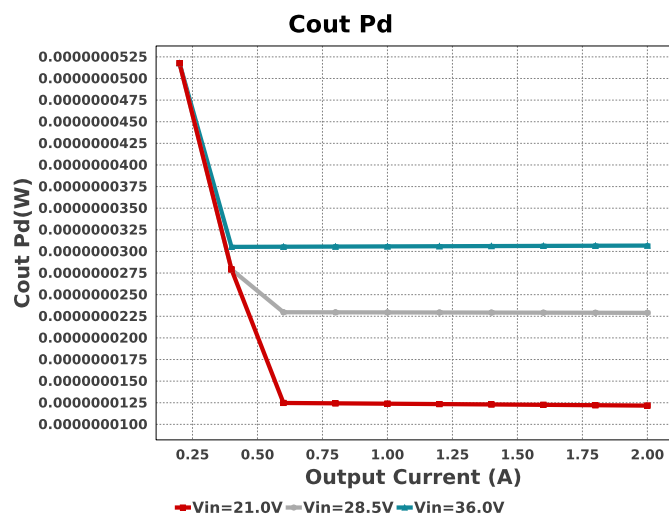
Electrical BOM

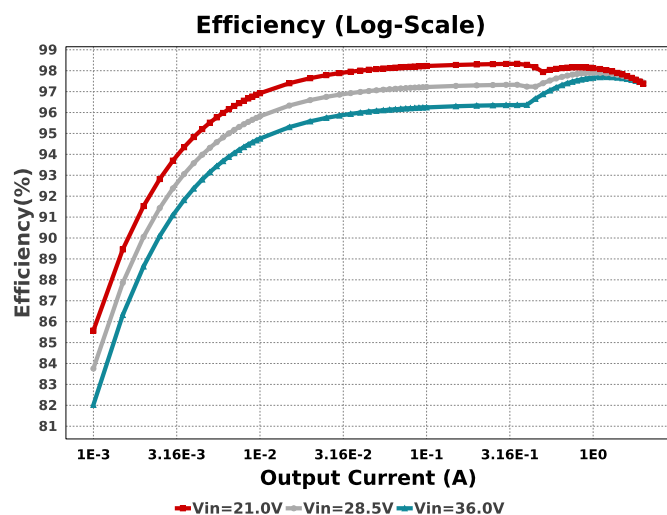
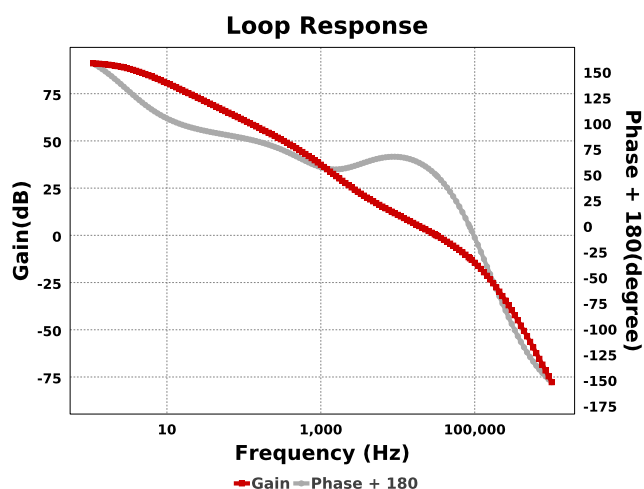
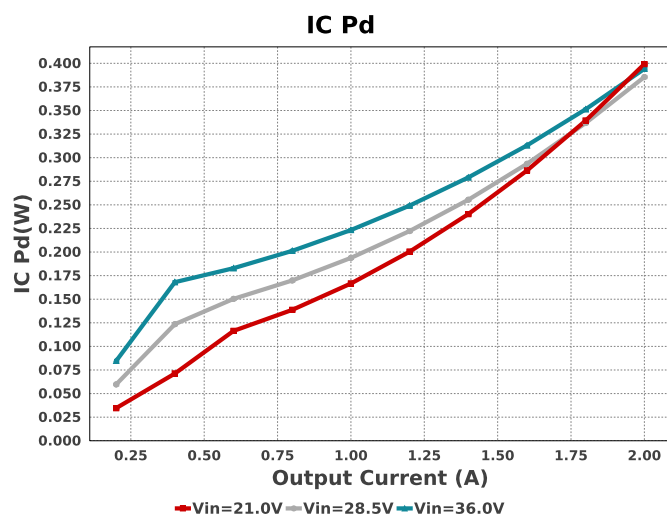
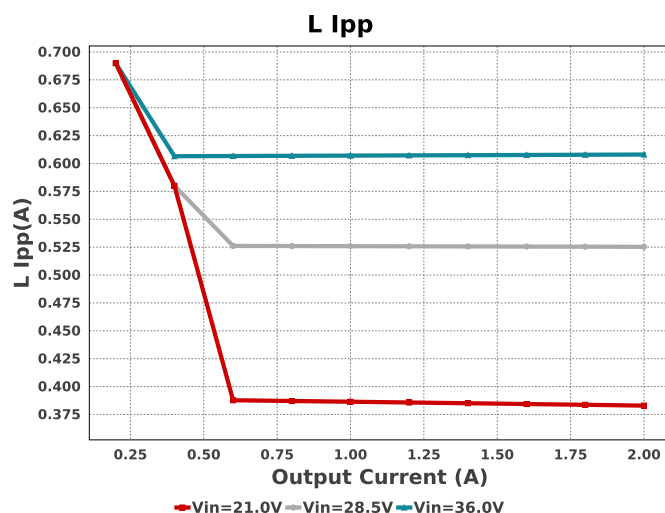
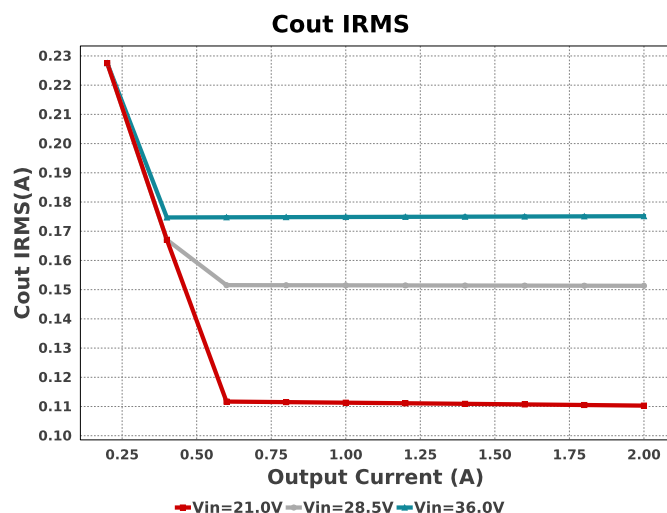
Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Cboot	MuRata	GRM155R71C104KA88D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 16.0 V IRMS= 0.0 A	1	\$0.01	0402 3 mm ²
Cin	MuRata	GRM31CR71H475KA12L Series= X7R	Cap= 4.7 uF ESR= 3.0 mOhm VDC= 50.0 V IRMS= 4.98 A	1	\$0.10	1206 11 mm ²
Cin_ch1	MuRata	GRM31CR71H475KA12L Series= X7R	Cap= 4.7 uF ESR= 3.0 mOhm VDC= 50.0 V IRMS= 4.98 A	1	\$0.10	1206 11 mm ²
Cinx	MuRata	GRM188R72A104KA35D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 100.0 V IRMS= 3.85 A	1	\$0.04	0603 5 mm ²
Cinx_ch1	MuRata	GRM188R72A104KA35D Series= X7R	Cap= 100.0 nF ESR= 1.0 mOhm VDC= 100.0 V IRMS= 3.85 A	1	\$0.04	0603 5 mm ²
Cout	CUSTOM	CUSTOM Series= ?	Cap= 44.0 uF ESR= 1.0 uOhm VDC= 17.1429 V IRMS= 250.742 mA	1	NA	CUSTOM 0 mm ²
Coutx	TDK	CGA3E2X7R1H104K080AA Series= X7R	Cap= 100.0 nF ESR= 29.6 mOhm VDC= 50.0 V IRMS= 971.99 mA	1	\$0.01	0603 5 mm ²
Cvcc	Kemet	C0603C105K8PACTU Series= X5R	Cap= 1.0 uF ESR= 1.0 mOhm VDC= 10.0 V IRMS= 0.0 A	1	\$0.01	0603 5 mm ²
L1	Würth Elektronik	784373865330	L= 33.0 uH 60.0 mOhm	1	\$2.19	WE-LHCA_1365 0 mm ²
Rfbb	Yageo	RC0603FR-079K09L Series= ?	Res= 9.09 kOhm Power= 100.0 mW Tolerance= 1.0%	1	\$0.01	0603 5 mm ²
Rfht	Vishay-Dale	CRCW0402100KFKED Series= CRCW..e3	Res= 100.0 kOhm Power= 63.0 mW Tolerance= 1.0%	1	\$0.01	0402 3 mm ²

Name	Manufacturer	Part Number	Properties	Qty	Price	Footprint
Rpg	Yageo	RC1206FR-07100KL Series= ?	Res= 100.0 kOhm Power= 250.0 mW Tolerance= 1.0%	1	\$0.01	 1206 11 mm²
U1	Texas Instruments	LMR33630ARNXR	Switcher	1	\$1.09	RNX0012B 12 mm²









Operating Values

#	Name	Value	Category	Description
1.	BOM Count	13		Total Design BOM count
2.	Total BOM	NA		Total BOM Cost
3.	Cin IRMS	918.937 mA	Capacitor	Input capacitor RMS ripple current
4.	Cin Pd	2.533 mW	Capacitor	Input capacitor power dissipation
5.	Cinx IRMS	33.484 mA	Capacitor	Bulk capacitor RMS ripple current
6.	Cinx Pd	1.121 μ W	Capacitor	Bulk capacitor power dissipation
7.	Cout IRMS	175.147 mA	Capacitor	Output capacitor RMS ripple current
8.	Cout Pd	30.676 nW	Capacitor	Output capacitor power dissipation
9.	Coutx IRMS	372.444 μ A	Capacitor	Output capacitor_x RMS ripple current
10.	Coutx Pd	4.106 nW	Capacitor	Output capacitor_x power loss
11.	IC Ipk	2.304 A	IC	Peak switch current in IC

#	Name	Value	Category	Description
12.	IC Pd	394.16 mW	IC	IC power dissipation
13.	IC Tj	45.766 degC	IC	IC junction temperature
14.	IC Tolerance	15.0 mV	IC	IC Feedback Tolerance
15.	ICThetaJA Effective	40.0 degC/W	IC	Effective IC Junction-to-Ambient Thermal Resistance
16.	Iin Avg	684.44 mA	IC	Average input current
17.	Ipp percentage	30.401 %	Inductor	Inductor ripple current percentage (with respect to average inductor current)
18.	L Ipp	608.016 mA	Inductor	Peak-to-peak inductor ripple current
19.	L Pd	241.85 mW	Inductor	Inductor power dissipation
20.	Cin Pd	2.533 mW	Power	Input capacitor power dissipation
21.	Cinx Pd	1.121 μ W	Power	Bulk capacitor power dissipation
22.	Cout Pd	30.676 nW	Power	Output capacitor power dissipation
23.	Coutx Pd	4.106 nW	Power	Output capacitor_x power loss
24.	IC Pd	394.16 mW	Power	IC power dissipation
25.	L Pd	241.85 mW	Power	Inductor power dissipation
26.	Total Pd	639.897 mW	Power	Total Power Dissipation
27.	Cross Freq	31.706 kHz	System Information	Bode plot crossover frequency
28.	Duty Cycle	33.926 %	System Information	Duty cycle
29.	Efficiency	97.403 %	System Information	Steady state efficiency
30.	FootPrint	174.0 mm ²	System Information	Total Foot Print Area of BOM components
31.	Frequency	400.0 kHz	System Information	Switching frequency
32.	Gain Marg	-12.202 dB	System Information	Bode Plot Gain Margin
33.	Iout	2.0 A	System Information	Iout operating point
34.	Low Freq Gain	91.073 dB	System Information	Gain at 1Hz
35.	Mode	CCM	System Information	Conduction Mode
36.	Phase Marg	50.475 deg	System Information	Bode Plot Phase Margin
37.	Pout	24.0 W	System Information	Total output power
38.	Vin	36.0 V	System Information	Vin operating point
39.	Vin p-p	622.168 mV	System Information	Peak-to-peak input voltage
40.	Vout	12.0 V	System Information	Operational Output Voltage
41.	Vout Actual	12.001 V	System Information	Vout Actual calculated based on selected voltage divider resistors
42.	Vout Tolerance	3.38 %	System Information	Vout Tolerance based on IC Tolerance (no load) and voltage divider resistors if applicable
43.	Vout p-p	4.309 mV	System Information	Peak-to-peak output ripple voltage

Design Inputs

Name	Value	Description
Iout	2.0	Maximum Output Current
VinMax	36.0	Maximum input voltage
VinMin	21.0	Minimum input voltage
Vout	12.0	Output Voltage
base_pn	LMR33630A-WS0N	Base Product Number
source	DC	Input Source Type
Ta	30.0	Ambient temperature

WEBENCH® Assembly

Component Testing

Some published data on components in datasheets such as Capacitor ESR and Inductor DC resistance is based on conservative values that will guarantee that the components always exceed the specification. For design purposes it is usually better to work with typical values. Since this data is not always available it is a good practice to measure the Capacitance and ESR values of C_{in} and C_{out} , and the inductance and DC resistance of $L1$ before assembly of the board. Any large discrepancies in values should be electrically simulated in WEBENCH to check for instabilities and thermally simulated in WebTHERM to make sure critical temperatures are not exceeded.

Soldering Component to Board

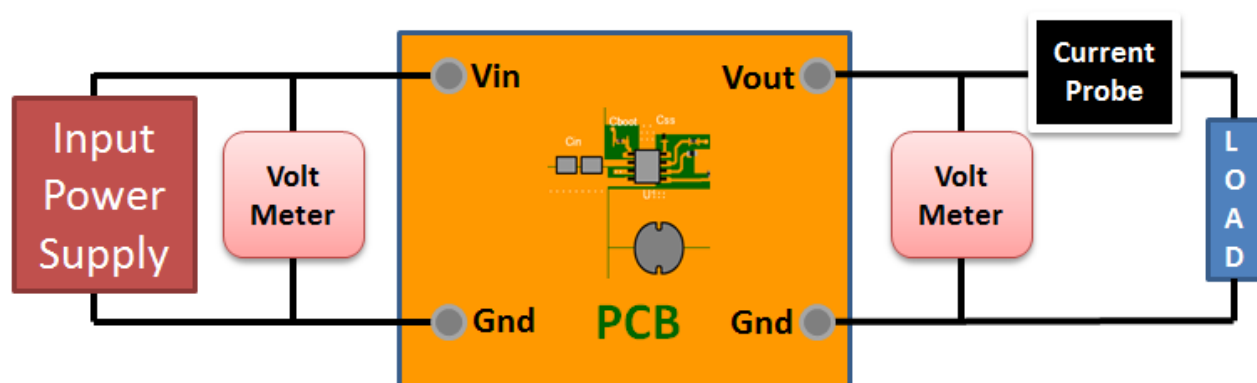
If board assembly is done in house it is best to tack down one terminal of a component on the board then solder the other terminal. For surface mount parts with large tabs, such as the DPAK, the tab on the back of the package should be pre-tinned with solder, then tacked into place by one of the pins. To solder the tab down to the board place the iron down on the board while resting against the tab, heating both surfaces simultaneously. Apply light pressure to the top of the plastic case until the solder flows around the part and the part is flush with the PCB. If the solder is not flowing around the board you may need a higher wattage iron (generally 25W to 30W is enough).

Initial Startup of Circuit

It is best to initially power up the board by setting the input supply voltage to the lowest operating input voltage 21.0V and set the input supply's current limit to zero. With the input supply off connect up the input supply to V_{in} and GND. Connect a digital volt meter and a load if needed to set the minimum load of the design from V_{out} and GND. Turn on the input supply and slowly turn up the current limit on the input supply. If the voltage starts to rise on the input supply continue increasing the input supply current limit while watching the output voltage. If the current increases on the input supply, but the voltage remains near zero, then there may be a short or a component misplaced on the board. Power down the board and visually inspect for solder bridges and recheck the diode and capacitor polarities. Once the power supply circuit is operational then more extensive testing may include full load testing, transient load and line tests to compare with simulation results.

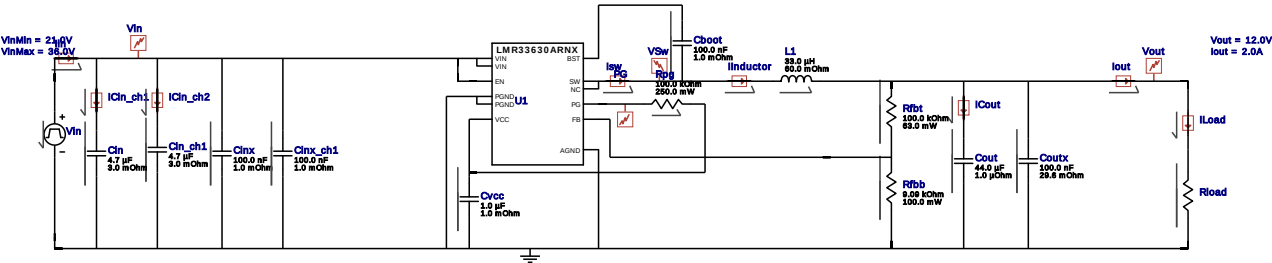
Load Testing

The setup is the same as the initial startup, except that an additional digital voltmeter is connected between V_{in} and GND, a load is connected between V_{out} and GND and a current meter is connected in series between V_{out} and the load. The load must be able to handle at least rated output power + 50% (7.5 watts for this design). Ideally the load is supplied in the form of a variable load test unit. It can also be done in the form of suitably large power resistors. When using an oscilloscope to measure waveforms on the prototype board, the ground leads of the oscilloscope probes should be as short as possible and the area of the loop formed by the ground lead should be kept to a minimum. This will help reduce ground lead inductance and eliminate EMI noise that is not actually present in the circuit.



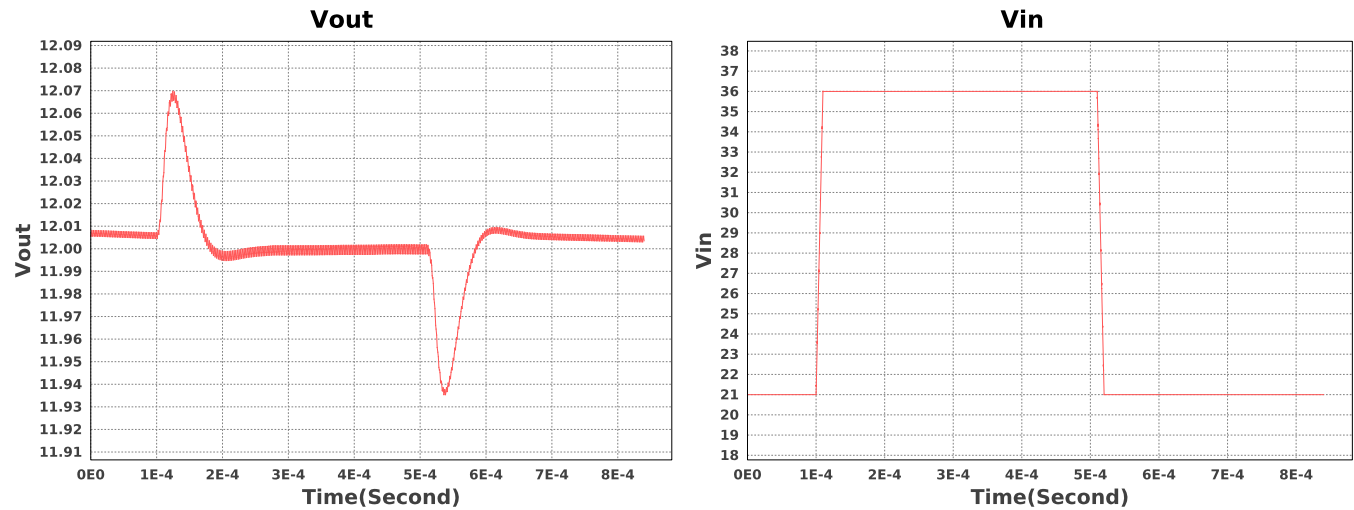
WEBENCH® Electrical Simulation Report

Design Id = 36
sim_id = 2
Simulation Type = Input Transient



Simulation Parameters

#	Name	Parameter Name	Description	Values
1.	Rload	R	Load Resistance	6.0 Ohm



Design Assistance

- Master key : 5FFB4611713714A73A2F511ACDE4AFEF[v1]
- LMR33630A-WSON Product Folder : <http://www.ti.com/product/LMR33630> : contains the data sheet and other resources.

Important Notice and Disclaimer

TI provides technical and reliability data (including datasheets), design resources (including reference designs), application or other design advice, web tools, safety information, and other resources AS IS and with all faults, and disclaims all warranties. These resources are intended for skilled developers designing with TI products. You are solely responsible for (1) selecting the appropriate TI products for your application, (2) designing, validating and testing your application, and (3) ensuring your application meets applicable standards, and any other safety, security, or other requirements.

These resources are subject to change without notice. TI grants you permission to use these resources only for development of an application that uses the TI products described in the resource. Other reproduction and display of these resources is prohibited. No license is granted to any other TI intellectual property right or to any third party intellectual property right. TI disclaims responsibility for, and you will fully indemnify TI and its representatives against, any claims, damages, costs, losses, and liabilities arising out of your use of these resources.

Providing these resources does not expand or otherwise alter TI's applicable Terms of Sale or other applicable terms available either on ti.com or provided in conjunction with TI products.